From Potentials and Chemicals to the Mind

The Neurological and Psychological Foundations of the Mind

Jack Wetherell

Contents

1 Nervous System Structure **2**
- 1.1 The Cerebrum . . . 2
- 1.2 The Cerebellum . . . 2
- 1.3 The Midbrain and Brainstem . . . 2
- 1.4 Cranial Nerves . . . 3
- 1.5 Ventricles . . . 3
- 1.6 Spinal Cord . . . 3
- 1.7 Peripheral Nerves . . . 4
- 1.8 Summary . . . 4

2 Neuron Structure **4**
- 2.1 Soma, Dendrites, and the Axon . . . 4
  - 2.1.1 Soma (Cell Body) . . . 4
  - 2.1.2 Dendrites . . . 5
  - 2.1.3 Axon . . . 5
- 2.2 The Nucleus, Rough Endoplasmic Reticulum (RER), and Golgi Apparatus . . . 5
  - 2.2.1 Nucleus . . . 5
  - 2.2.2 Rough Endoplasmic Reticulum (RER) . . . 5
  - 2.2.3 Golgi Apparatus . . . 6
- 2.3 Microtubules, Kinesin, and Dynein . . . 6
  - 2.3.1 Microtubules . . . 6
  - 2.3.2 Kinesin and Dynein . . . 6
- 2.4 Vesicles and SNARE Proteins . . . 6
  - 2.4.1 Synaptic Vesicles . . . 6
  - 2.4.2 SNARE Proteins . . . 7
- 2.5 Neurotransmitters . . . 7
- 2.6 Summary . . . 7

3 Neuron Firing **7**
- 3.1 Resting Membrane Potential . . . 8
  - 3.1.1 Nernst Equation . . . 8
  - 3.1.2 Typical Ion Equilibrium Potentials . . . 8
- 3.2 Graded Potentials . . . 8
- 3.3 Action Potentials . . . 8
  - 3.3.1 Threshold . . . 9
  - 3.3.2 Phases of the Action Potential . . . 9
- 3.4 Neurotransmitter Release . . . 9
- 3.5 Reuptake . . . 10
- 3.6 Refractory Periods . . . 10

3.6.1 Absolute Refractory Period 10
3.6.2 Relative Refractory Period 10
3.7 Summary 10

1 Nervous System Structure

The nervous system is a highly organized and complex network responsible for coordinating the body's activities, processing sensory information, and enabling cognition, movement, and homeostasis. It is broadly divided into two main components:

- **Central Nervous System (CNS)**: Comprises the brain and spinal cord. It is responsible for integrating information and generating responses.
- **Peripheral Nervous System (PNS)**: Consists of nerves outside the CNS and connects the CNS to the rest of the body.

This section focuses on the major anatomical structures of the nervous system and their functions.

1.1 The Cerebrum

The cerebrum is the largest part of the brain and is responsible for higher cognitive functions such as thinking, reasoning, memory, and voluntary movement.

It is divided into two hemispheres (left and right), connected by the corpus callosum, which allows communication between them. Each hemisphere is further divided into four lobes:

- **Frontal lobe**: Involved in decision-making, planning, personality, and voluntary motor control.
- **Parietal lobe**: Processes sensory information such as touch, temperature, and pain.
- **Temporal lobe**: Important for hearing, language comprehension, and memory.
- **Occipital lobe**: Responsible for visual processing.

The outer layer of the cerebrum, known as the **cerebral cortex**, is highly folded, increasing surface area and allowing for greater cognitive capacity.

1.2 The Cerebellum

The cerebellum is located beneath the cerebrum and plays a crucial role in coordinating movement, balance, and posture.

Although it does not initiate movement, it fine-tunes motor activity by comparing intended movement with actual performance. This allows for smooth, precise, and coordinated actions.

Damage to the cerebellum can result in:

- Ataxia (loss of coordination)
- Tremors
- Difficulty maintaining balance

1.3 The Midbrain and Brainstem

The brainstem connects the brain to the spinal cord and is essential for basic life functions.

It is divided into three parts:

- **Midbrain**: Involved in visual and auditory reflexes and motor control.
- **Pons**: Acts as a relay station between different parts of the brain and is involved in sleep and respiration.

- **Medulla oblongata**: Regulates vital autonomic functions such as heart rate, breathing, and blood pressure.

The brainstem is critical for survival, as it controls involuntary processes necessary for life.

1.4 Cranial Nerves

Cranial nerves are a set of 12 paired nerves that emerge directly from the brain rather than the spinal cord.

They are responsible for sensory and motor functions of the head and neck, including:

- Vision (optic nerve)
- Eye movement (oculomotor, trochlear, abducens nerves)
- Facial sensation and expression (trigeminal and facial nerves)
- Hearing and balance (vestibulocochlear nerve)
- Swallowing and speech (glossopharyngeal and vagus nerves)

Each cranial nerve has a specific function and can be tested clinically to assess neurological function.

1.5 Ventricles

The ventricles are interconnected cavities within the brain filled with cerebrospinal fluid (CSF).

There are four ventricles:

- Two lateral ventricles
- Third ventricle
- Fourth ventricle

CSF serves several important functions:

- Cushions and protects the brain
- Maintains chemical stability
- Removes waste products

CSF is produced by the choroid plexus and circulates through the ventricles and around the brain and spinal cord.

1.6 Spinal Cord

The spinal cord is a long, cylindrical structure that extends from the brainstem down the vertebral column.

It serves two main functions:

- **Transmission**: Carries sensory information to the brain and motor commands from the brain to the body.
- **Reflexes**: Mediates rapid, automatic responses to stimuli without requiring conscious input from the brain.

The spinal cord is segmented, with each segment giving rise to a pair of spinal nerves.

1.7 Peripheral Nerves

Peripheral nerves are part of the PNS and connect the CNS to muscles, glands, and sensory organs.

They are classified into:

- **Sensory (afferent) nerves**: Carry information from the body to the CNS.
- **Motor (efferent) nerves**: Carry commands from the CNS to muscles and glands.

The PNS can also be divided into:

- **Somatic nervous system**: Controls voluntary movements.
- **Autonomic nervous system**: Regulates involuntary functions such as heart rate and digestion.

The autonomic nervous system is further subdivided into:

- **Sympathetic division**: Prepares the body for “fight or flight” responses.
- **Parasympathetic division**: Promotes “rest and digest” functions.

1.8 Summary

The nervous system is a hierarchical and integrated system in which different structures work together to process information and coordinate responses. The brain handles complex processing, the spinal cord relays and integrates signals, and peripheral nerves connect the entire system to the body.

Understanding these structural components provides the foundation for studying neuronal function, signaling, and neurological disorders.

2 Neuron Structure

Neurons are the fundamental functional units of the nervous system. They are specialized cells designed to transmit electrical and chemical signals over long distances, enabling communication within the body. Unlike most other cell types, neurons are highly polarized, meaning their structure is asymmetrical and optimized for directional signaling.

A typical neuron consists of three main regions: the soma (cell body), dendrites, and the axon. In addition, neurons contain specialized intracellular structures that support their high metabolic and signaling demands.

2.1 Soma, Dendrites, and the Axon

2.1.1 Soma (Cell Body)

The soma is the central part of the neuron and contains the nucleus and most of the cell’s organelles. It serves as the metabolic and synthetic center of the neuron.

Key functions of the soma include:

- Integration of incoming signals from dendrites
- Synthesis of proteins and neurotransmitters
- Maintenance of cellular health and homeostasis

The soma plays a critical role in determining whether a neuron will generate an action potential, based on the inputs it receives.

2.1.2 Dendrites

Dendrites are branching extensions that arise from the soma. They are the primary sites for receiving synaptic input from other neurons.

Key features of dendrites:

- Highly branched to increase surface area
- Contain dendritic spines, which are small protrusions where synapses form
- Specialized for receiving chemical signals and converting them into electrical signals (graded potentials)

The structure of dendrites allows neurons to receive input from thousands of other neurons simultaneously.

2.1.3 Axon

The axon is a long, thin projection that carries electrical signals away from the soma toward other neurons or target cells (e.g., muscle or gland cells).

Important regions of the axon include:

- **Axon hillock**: The region where the axon emerges from the soma; this is where action potentials are typically initiated.
- **Axon shaft**: Conducts the action potential along its length.
- **Axon terminals (synaptic boutons)**: The distal ends where neurotransmitters are released.

Some axons are covered in myelin, a fatty insulating layer that increases the speed of signal transmission.

2.2 The Nucleus, Rough Endoplasmic Reticulum (RER), and Golgi Apparatus

Neurons have high metabolic demands and require efficient systems for protein synthesis and transport.

2.2.1 Nucleus

The nucleus contains the cell's genetic material (DNA) and controls gene expression. It regulates the production of proteins necessary for neuronal function, including receptors, ion channels, and structural proteins.

2.2.2 Rough Endoplasmic Reticulum (RER)

In neurons, the RER is highly developed and is referred to as **Nissl substance**. It is responsible for synthesizing proteins, particularly those destined for membranes or secretion.

Functions include:

- Production of neurotransmitter-related proteins
- Synthesis of ion channels and receptors

2.2.3 Golgi Apparatus

The Golgi apparatus modifies, sorts, and packages proteins produced by the RER.

Its roles include:

- Post-translational modification of proteins
- Packaging proteins into vesicles
- Directing proteins to their appropriate cellular destinations

This system ensures that proteins are correctly processed and delivered where needed within the neuron.

2.3 Microtubules, Kinesin, and Dynein

Neurons rely heavily on intracellular transport due to their length, especially in long axons.

2.3.1 Microtubules

Microtubules are structural components of the cytoskeleton that act as tracks for intracellular transport. They extend throughout the axon and dendrites.

Functions:

- Provide structural support
- Serve as pathways for transporting organelles and vesicles

2.3.2 Kinesin and Dynein

These are motor proteins that move along microtubules, carrying cargo within the neuron.

- **Kinesin**: Moves cargo away from the soma (anterograde transport), toward the axon terminals.
- **Dynein**: Moves cargo toward the soma (retrograde transport).

This bidirectional transport system is essential for:

- Delivering neurotransmitters and proteins to synapses
- Recycling materials from axon terminals
- Maintaining neuronal health

2.4 Vesicles and SNARE Proteins

2.4.1 Synaptic Vesicles

Synaptic vesicles are small membrane-bound structures found in axon terminals. They store neurotransmitters and release them into the synaptic cleft during neuronal communication.

Key steps:

- Vesicles are filled with neurotransmitters
- They move toward the presynaptic membrane
- They fuse with the membrane and release contents via exocytosis

2.4.2 SNARE Proteins

SNARE proteins are a group of proteins that mediate the fusion of vesicles with the presynaptic membrane.

They ensure:

- Precise docking of vesicles
- Rapid and controlled neurotransmitter release

This process is tightly regulated and often triggered by calcium influx during an action potential.

2.5 Neurotransmitters

Neurotransmitters are chemical messengers that transmit signals across synapses from one neuron to another.

They can be broadly classified into:

- **Excitatory neurotransmitters** (e.g., glutamate): Increase the likelihood of an action potential.
- **Inhibitory neurotransmitters** (e.g., GABA): Decrease the likelihood of an action potential.

Other important neurotransmitters include:

- Dopamine (movement, reward)
- Serotonin (mood, sleep)
- Acetylcholine (muscle activation, memory)

Neurotransmitters bind to receptors on the postsynaptic cell, leading to changes in electrical activity.

2.6 Summary

Neuron structure is highly specialized to support rapid and efficient communication. The soma integrates signals, dendrites receive input, and the axon transmits output. Intracellular systems such as the RER, Golgi apparatus, and cytoskeleton ensure that proteins and materials are produced and transported effectively. Vesicles and neurotransmitters enable communication between neurons, forming the basis of all nervous system function.

Understanding neuronal structure is essential for studying how neurons generate and transmit signals, which is explored in the next section on neuron firing.

3 Neuron Firing

Neuron firing refers to the process by which neurons generate and transmit electrical signals. This involves changes in the membrane potential due to the movement of ions across the neuronal membrane. These electrical signals allow neurons to communicate rapidly and efficiently.

3.1 Resting Membrane Potential

The resting membrane potential is the electrical potential difference across the membrane of a neuron when it is not actively firing. It is typically around:

$$V_m \approx -70\,\mathrm{mV}$$

This negative potential arises due to:

- Unequal distribution of ions (Na^+, K^+, Cl^-)
- Selective permeability of the membrane (more permeable to K^+)
- The activity of the Na^+/K^+ ATPase pump (3 Na^+ out, 2 K^+ in)

3.1.1 Nernst Equation

The equilibrium potential for a single ion can be calculated using the Nernst equation:
:contentReference[oaicite:0]index=0
At body temperature (37°C), this simplifies to:

$$E_{ion} \approx \frac{61}{z} \log \left(\frac{[ion]_{out}}{[ion]_{in}} \right) \text{ mV}$$

3.1.2 Typical Ion Equilibrium Potentials

- Potassium (K^+): ≈ -90 mV
- Sodium (Na^+): $\approx +60$ mV
- Chloride (Cl^-): ≈ -65 mV

The resting membrane potential is closest to the potassium equilibrium potential because the membrane is most permeable to K^+ at rest.

3.2 Graded Potentials

Graded potentials are small changes in membrane potential that occur in dendrites and the soma.

Characteristics:

- Can be depolarizing (less negative) or hyperpolarizing (more negative)
- Decrease in strength with distance (not all-or-none)
- Summate spatially and temporally

Examples:

- Excitatory postsynaptic potentials (EPSPs): move membrane toward threshold
- Inhibitory postsynaptic potentials (IPSPs): move membrane away from threshold

3.3 Action Potentials

An action potential is a rapid, all-or-none electrical signal that travels along the axon.

3.3.1 Threshold

If the membrane potential reaches approximately:

$$-55\,\text{mV}$$

voltage-gated Na^+ channels open, initiating the action potential.

3.3.2 Phases of the Action Potential

1. Resting State (≈ -70 mV)

- Voltage-gated channels closed
- Leak channels maintain resting potential

2. Depolarization (up to $\approx +30$ mV)

- Rapid opening of voltage-gated Na^+ channels
- Na^+ influx drives membrane potential toward E_{Na} ($\approx +60$ mV)

3. Repolarization

- Na^+ channels inactivate
- Voltage-gated K^+ channels open
- K^+ exits the cell

4. Hyperpolarization (≈ -80 to -90 mV)

- Continued K^+ efflux
- Membrane approaches E_K

5. Return to Rest

- K^+ channels close
- Na^+/K^+ pump restores ion gradients

3.4 Neurotransmitter Release

When an action potential reaches the axon terminal:

- Voltage-gated Ca^{2+} channels open
- Ca^{2+} enters the terminal
- This triggers vesicle fusion via SNARE proteins
- Neurotransmitters are released into the synaptic cleft

The amount of neurotransmitter released is proportional to calcium influx.

3.5 Reuptake

After release, neurotransmitters must be cleared from the synapse to terminate the signal.
Mechanisms include:

- **Reuptake**: Transport back into the presynaptic neuron
- **Enzymatic degradation**: e.g., acetylcholine broken down by acetylcholinesterase
- **Diffusion**: Away from the synaptic cleft

Reuptake is a key target for many drugs (e.g., SSRIs for serotonin).

3.6 Refractory Periods

After an action potential, the neuron enters a refractory period:

3.6.1 Absolute Refractory Period

- No new action potential can be generated
- Due to inactivation of Na^+ channels

3.6.2 Relative Refractory Period

- A stronger-than-normal stimulus is required
- Occurs during hyperpolarization

These periods ensure:

- Unidirectional propagation of action potentials
- Controlled firing frequency

3.7 Summary

Neuron firing is a tightly regulated process involving ion movement across the membrane. The resting membrane potential is primarily determined by potassium permeability, while action potentials arise from rapid changes in sodium and potassium conductance. The Nernst equation provides a quantitative understanding of ion equilibrium potentials, which underpin neuronal excitability. Neurotransmitter release and reuptake enable communication between neurons, while refractory periods ensure proper signal timing and direction.